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PRN NO-

202501040081BATCH-

CS1

PRACTICAL NO.1

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum is calculated using the formula:

where:

is the mass of the object (in kilograms).

is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

CODE:

```
m = float(input())
v = float(input())
p = m*v
print(f"{p:.2f}kgm/s")
```

OUTPUT:



Write a Python program that accepts an integer as input. Depending on the number of digits in .

Constraints:

$$1 \leq \leq 999$$

Input Format:

- The input consists of a single integer .

Output Format:

- If is a single-digit number, print its square.
- If is a two-digit number, print its square root (rounded to two decimal places).
- If is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

CODE:

```
n = int(input())
```

```
if 1<=n<=9:
```

```
    print(n**2)
```

elif 10<=n<=99:

```
    print(f"{n**0.5:.2f}")
```

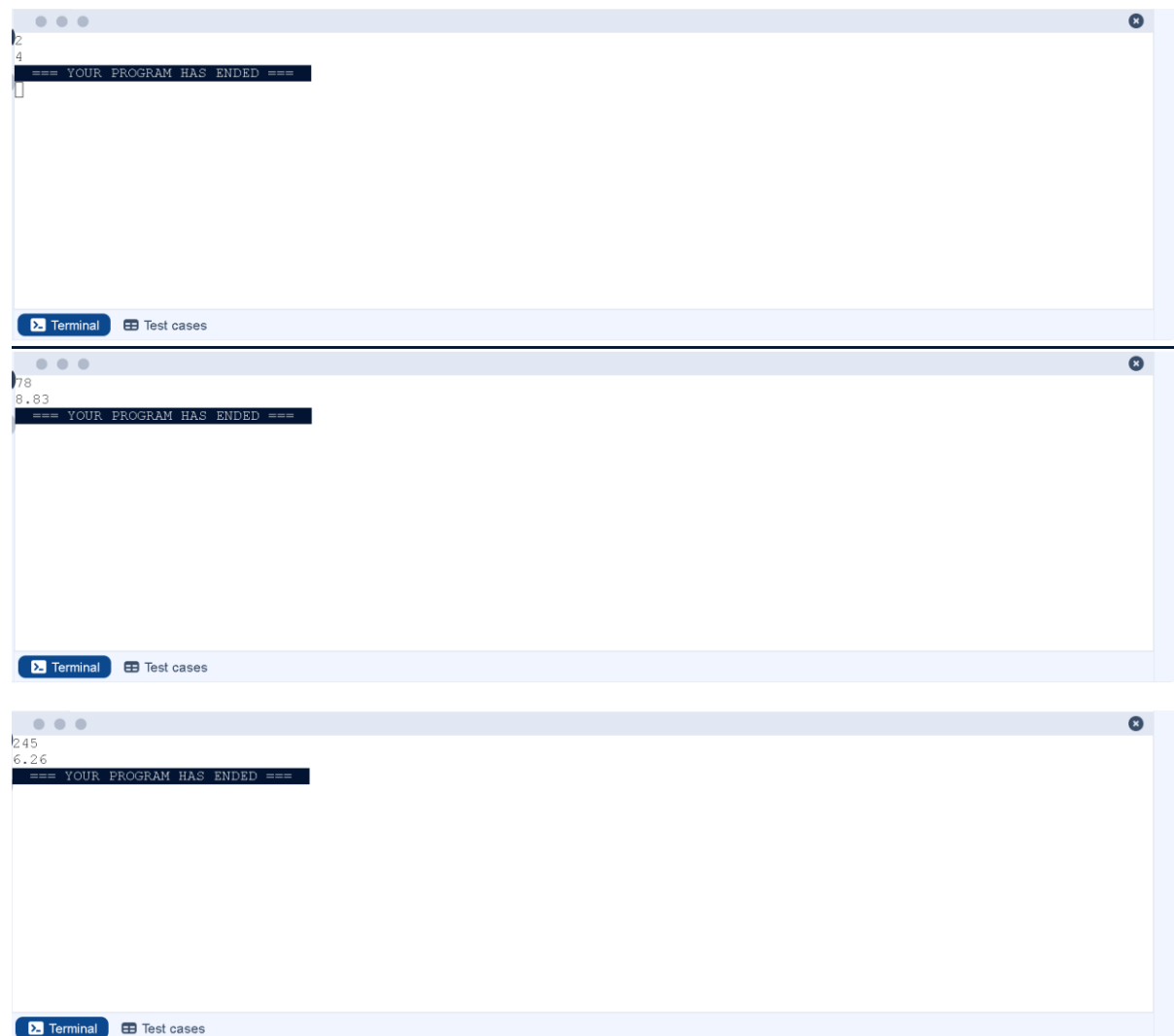
elif 100<=n<=999:

```
    print(f"{n**(1/3):.2f}")
```

else:

```
    print("Invalid")
```

OUTPUT:



Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format .YYYY-MM-DD
- The second line contains the second date in the format .YYYY-MM-DD

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

CODE:

```
from datetime import datetime
date1 = input().strip()
date2 = input().strip()

d1 = datetime.strptime(date1, "%Y-%m-%d")
d2 = datetime.strptime(date2, "%Y-%m-%d")
difference = d2-d1
print(difference.days)
```

OUTPUT:



Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

CODE:

```
n = int(input())
```

```
rev = 0
```

```
while n>0:
```

```
    digit = n%10
```

```
    r
```

```
    e
```

```
    v
```

```
    =
```

```
    r
```

```
    e
```

```
    v
```

```
    *
```

```
    1
```

```
    0
```

```
    +
```

```
    d
```

```
    i
```

```
    g
```

```
    i
```

t /

n /

= 1

n 0

print(rev)

OUTPUT:



```
123456
654321
=== YOUR PROGRAM HAS ENDED ===
```

The image shows a terminal window with a light blue header bar containing three window control buttons on the left and a close button on the right. The terminal area has a white background. The output consists of two lines of numbers, '123456' and '654321', followed by a dark blue rectangular box containing the text '=== YOUR PROGRAM HAS ENDED ===' in white. Below the terminal area is a dark blue footer bar with two tabs: 'Terminal' (active, with a terminal icon) and 'Test cases' (inactive, with a document icon).

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:

<number> x <multiplier> = <result>

CODE:

```
n = int(input())
```

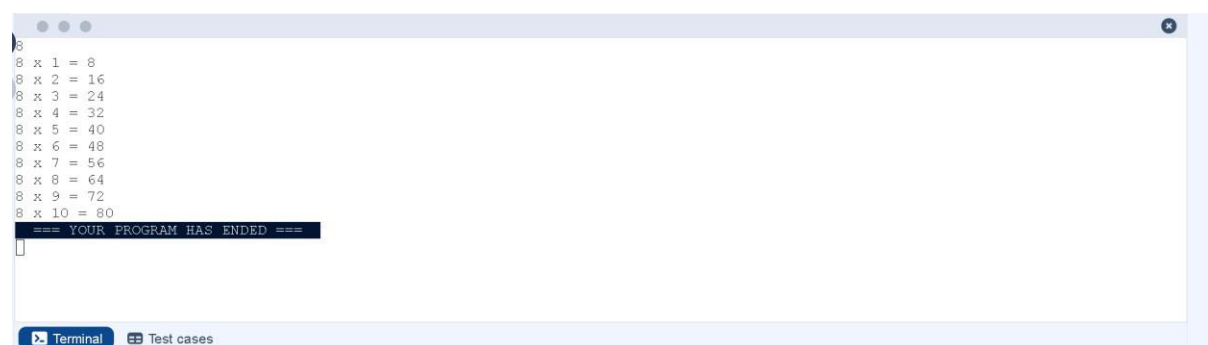
```
for i in  
range(1,11):
```

print

```
(n, "x", i,  
"=", n*i)
```

OUTPUT:

T:



```
8
8 x 1 = 8
8 x 2 = 16
8 x 3 = 24
8 x 4 = 32
8 x 5 = 40
8 x 6 = 48
8 x 7 = 56
8 x 8 = 64
8 x 9 = 72
8 x 10 = 80
=== YOUR PROGRAM HAS ENDED ===
```

Write a Python program that accepts the number of courses and the marks of a student in those courses.

The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer , the number of courses.
- The second input will be integers representing the marks of the student in each of the courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

CODE:

```
n = int(input())
```

```
marks = list(map(int, input().split()))
```

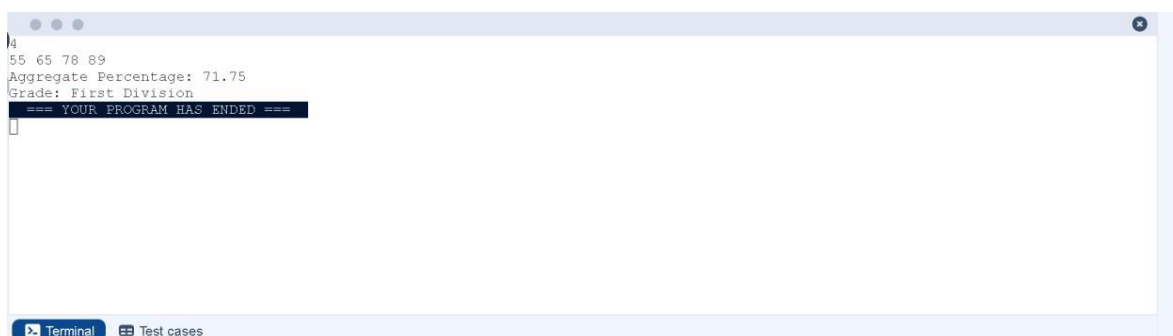
```
if any(m < 40 for m in marks):
```

```
    print("Fail")
else:
    aggregate = sum(marks) / n

    if aggregate > 75:
        grade =
        "Distinction"
    elif aggregate >= 60:
        grade = "First Division"
    elif aggregate >= 50:
        grade = "Second Division"
    elif aggregate >= 40:
        grade = "Third Division"

    print(f"Aggregate Percentage: {aggregate:.2f}")
    print(f"Grade: {grade}")
```

OUTPUT:

A screenshot of a terminal window with a light blue header bar containing window control buttons and a close button. The terminal area is white and displays the following text: a prompt character followed by the input '55 65 78 89', the output 'Aggregate Percentage: 71.75', the output 'Grade: First Division', and a blue banner with the text '=== YOUR PROGRAM HAS ENDED ==='. At the bottom, there is a dark blue bar with two buttons: 'Terminal' and 'Test cases'.

```
4
55 65 78 89
Aggregate Percentage: 71.75
Grade: First Division
=== YOUR PROGRAM HAS ENDED ===
```

Write a Python program that uses recursion to print the first terms of the Fibonacci series.

Input Format:

- A single integer representing the number of terms to generate.

Output Format:

- A single line containing the first terms of the Fibonacci sequence, separated by spaces.

CODE:

```
def fibonacci(n)
if n == 1:
return 0
elif n == 2:
return 1
else:
return fibonacci(n - 1) + fibonacci(n - 2)
n = int(input())
for i in range(1, n + 1):
print(fibonacci(i), end=" ")
```

OUTPUT:



```
5
0 1 1 2 3  === YOUR PROGRAM HAS ENDED ===
[]
```

The image shows a terminal window with a light blue header bar containing three window control buttons (minimize, maximize, close) on the left and a close button on the right. The main area of the terminal is white. The output of the program is displayed as follows: the first line is '5', the second line is '0 1 1 2 3', and the third line is '[]'. A dark blue rectangular box containing the text '=== YOUR PROGRAM HAS ENDED ===' is positioned to the right of the second line. At the bottom of the terminal window, there is a light blue bar with two buttons: 'Terminal' (with a terminal icon) and 'Test cases' (with a test case icon).

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format

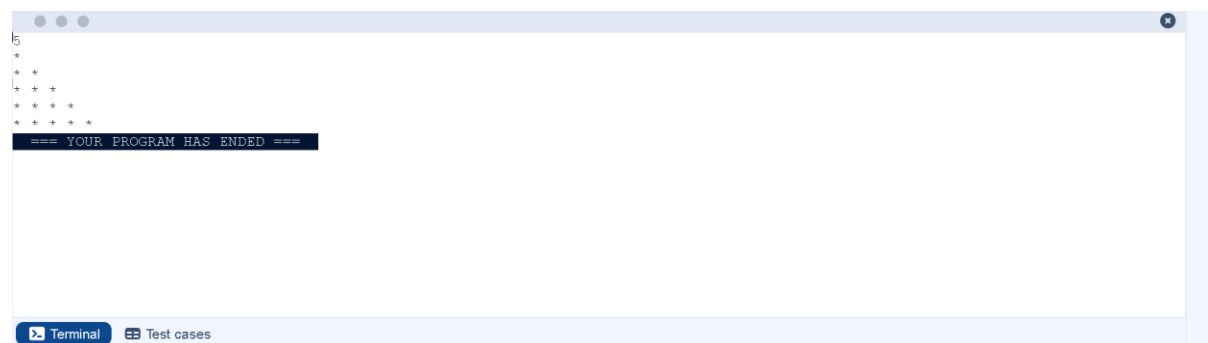
- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

CODE:

```
n = int(input())
for i in range(1, n+1):
    print("* " * i)
```

OUTPUT:



```
5
*
*
* *
* * *
* * * *
* * * * *
```

=== YOUR PROGRAM HAS ENDED ===

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

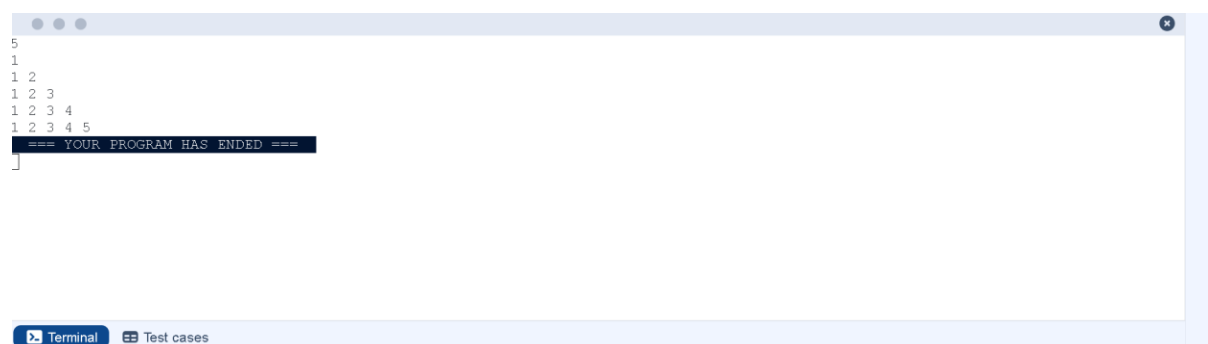
Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

CODE:

```
n = int(input())
for i in range(1, n+1):
    for j in range(1, i+1):
        print(j, end=" ")
    print()
```

OUTPUT:



```
5
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
=== YOUR PROGRAM HAS ENDED ===
```